

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and infrastructure Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTCIC402 - ENVIRONMENTAL ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Design population of a water supply scheme depends on:
A. Present population
B. Past census only
C. Population forecast
D. Daily demand
 - 2 Maximum daily demand is generally:
A. Equal to average demand
B. Less than average demand
C. 1.8 times average demand
D. Independent of average demand
 - 3 Main purpose of aeration is to remove:
A. Hardness
B. Suspended solids
C. Dissolved gases, Fe and Mn
D. Bacteria
 - 4 Rapid mixing is required mainly to:
A. Disperse coagulant uniformly
B. Settle particles
C. Form flocs
D. Remove odour
 - 5 If filtration rate is $5 \text{ m}^3/\text{m}^2 \cdot \text{h}$ and flow is $500 \text{ m}^3/\text{h}$, filter area required is:
A. 50 m^2
B. 75 m^2
C. 100 m^2
D. 125 m^2
 - 6 Major drawback of intermittent water supply is:
A. High cost
B. Low pressure
C. Risk of contamination
D. Leakage

- 7 Grid-iron system is suitable for:
 - A. Hilly areas
 - B. Circular cities
 - C. Well Planned cities
 - D. Rural areas
- 8 Hazen-Williams equation is used to calculate:
 - A. Velocity
 - B. Discharge
 - C. Head loss
 - D. Pressure
- 9 Main objective of wastewater treatment is to:
 - A. Improve colour
 - B. Save chemicals
 - C. Protect public health and enable safe disposal of treated wastewater
 - D. Increase flow
- 10 Temperature inversion results in:
 - A. Better dispersion
 - B. Vertical mixing
 - C. Stable atmosphere
 - D. Heavy rainfall
- 11 Most effective device for removal of particulate matter is:
 - A. Scrubber
 - B. ESP
 - C. Absorber
 - D. Cyclone only
- 12 Noise pollution is measured in:
 - A. dB
 - B. ppm
 - C. Hz
 - D. mg/m³

Q2. Solve the following.

- A) Enlist various population forecasting methods. Explain any one method. 6
- B) The population of a town as per census records is given below: 6

Year	Population
1991	50,000
2001	56,000
2011	62,000

Estimate the population for the year **2031** using the **Arithmetic Increase Method**.

Q3. Solve the following.

- A) Design a circular plain sedimentation tank for flow of 5MLD. Assume necessary data 6
- B) differentiate between Rapid sand filter and Slow sand filter 6

Q4. Solve Any Two of the following.

- A) Explain ring system of distribution network layout 6
- B) Explain Combined pumping and gravity system, How it is superior to other systems? 6
- C) Discuss tree system with neat sketch . Also list advantages and disadvantages 6

- Q5. Solve Any Two of the following.
- A) Classify solid waste based on sources. Give examples of each type. 6
 - B) Draw a neat flowsheet of sewage treatment plant. tabulate function of each and every unit. 6
 - C) Discuss Indore method of composting along with advantages and disadvantages. 6
- Q6. Solve Any Two of the following.
- A) Enlist types of air pollution control equipment. Discuss functioning of wet scrubber along with advantages and disadvantages. 6
 - B) Classify sources of air pollution in detail. 6
 - C) Define inversion. Classify stability conditions. 6

***** End *****